

48) Establish these logical equivalences, where x does not occur as a free variable in A . Assume that the domain is nonempty.

a) $(\forall x P(x)) \vee A \equiv \forall x (P(x) \vee A)$ Assume A has no free variables.

Let $(\forall x P(x)) \vee A$ be False. By defn. of OR, $\forall x P(x)$ is False and A is False. By defn. of negation, $\neg \forall x P(x)$ is True. By De Morgan's Law of quantifiers, $\exists x \neg P(x)$ is True. By defn. of existential quantifier, there exists an element a in the domain such that $\neg P(a)$ is True. By defn. of negation, $\neg(\neg P(a))$ is False. By Double-negation Law, $P(a)$ is False. $\therefore$ By defn. of OR, $P(a) \vee A$ is False. $\therefore$ By defn. of universal quantifier, $\forall x (P(x) \vee A)$ is False. $\therefore (\forall x P(x)) \vee A \Rightarrow \forall x (P(x) \vee A)$ is False

Let $\forall x (P(x) \vee A)$ be False. By defn. of negation, $\neg \forall x (P(x) \vee A)$ is True. By De Morgan's Law of quantifiers, $\exists x \neg (P(x) \vee A)$ is True. By 2nd De Morgan's Law, $\exists x (\neg P(x) \wedge \neg A)$ is True. By defn. of existential quantifier, there exists an element a in the domain such that $\neg P(a) \wedge \neg A$ is True. By defn. of AND, $\neg P(a)$ is True and $\neg A$ is True. By defn. of negation, $\neg(\neg P(a))$ is False and $\neg(\neg A)$ is False. By double negation law, $P(a)$ is False and A is False. By defn. of universal quantifier, $\forall x P(x)$ is False. By defn. of OR, $(\forall x P(x)) \vee A$ is False.

$\forall x (P(x) \vee A) \text{ is False} \Rightarrow (\forall x P(x)) \vee A \text{ is False}$

$\therefore$ They are logically equivalent $\square$

b) $(\exists x P(x)) \vee A \equiv \exists x (P(x) \vee A)$

Let $(\exists x P(x)) \vee A$ be False. By defn. of OR, $\exists x P(x)$ is False and A is False. By defn. of negation, $\neg \exists x P(x)$ is True. By de Morgan's Law of quantifiers, $\forall x \neg P(x)$ is True. By defn. of universal quantifier, for all elements a in the domain, $\neg P(a)$ is True. By defn. of negation,